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BUSINESS & INNOVATION**

NAME OF ASSIGNMENT : Enterprise Data Warehouses and Database Management Systems

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Management System**

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TABLE OF CONTENTS

- 1. Introduction
- 2. Task 1: Database Design
 - 2.1 Domain Description
 - 2.2 Entity Analysis
 - 2.3 Table Design
 - 2.4 ERD
 - 2.5 Normalization
- 3. Task 2: Sample Data and Views
- 4. Task 3: SQL Querying and Analytics
- 5. Task 4: CAP Theorem
- 6. Task 5: Query Performance Optimization
- 7. Conclusion
- 8. References
- 9. Appendices

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1 INTRODUCTION

In today's organizations, having an efficient database system to store, organize and manage data is vital. The librarians in the library use a database management system to keep track of books, members, their lenders, their loans and reservations. The proper designing of a database system can minimize data redundancy, enhance data integrity, and facilitate easy retrieval of information. This report aims to explore the designing and building of the relational library management system with the aid of Postgres in Supabase. The project gives an understanding about: Database design principles, Normalization, SQL implementation, Analytical query, CAP theorem, and Optimization of query performance.

2 Task 1 – Database Design

2.1 Domain Description

This project is about Library management system. Many books are stored in library and much information is stored about people using libraries, the staff, about loaning and reserving. The database includes information about books, books' category, borrowers and reservationers etc.

Main users of the system are:

- Library members
- Librarians
- Library administrators

It enables the monitoring of the availability of books, management of lending practices and useful analysis.

2.2 Entity Analysis

Chosen Domain: Library Management System Members, authors, books, categorization, librarians, loans and reservations data are saved. This includes library patrons, librarians and library administrators.

Entities:

- Members
- Authors
- Categories
- Books
- Librarians
- Loans
- BookReservations

SQL Editor

Members Authors Categories Books Librarians Loans BookReservations +

1 CREATE TABLE Members (
2 member_id INT PRIMARY KEY,
3 name VARCHAR(100),
4 email VARCHAR(100),
5 phone VARCHAR(20),
6 join_date DATE
7);
8

Results Explain Chart Export

Success. No rows returned

SQL Editor

Members Authors Categories Books Librarians Loans BookReservations +

1 CREATE TABLE Authors (
2 author_id INT PRIMARY KEY,
3 author_name VARCHAR(100),
4 country VARCHAR(50),
5 birth_year INT
6);

Results Explain Chart Export

Success. No rows returned

SQL Editor

Members Authors Categories Books Librarians Loans BookReservations List Public Tables

Search queries... +

SHARED

FAVORITES

PRIVATE (6)

- List Public Tables
- BookReservations
- Loans
- Librarians
- Books
- Categories
- Authors
- Members

REFERENCE

- Templates
- Examples

```
1 CREATE TABLE Categories (  
2   category_id INT PRIMARY KEY,  
3   category_name VARCHAR(50),  
4   description VARCHAR(200)  
5 );
```

Results Explain Chart Export

Success. No rows returned

SQL Editor

Members Authors Categories Books Librarians Loans BookReservations List Public Tables

Search queries... +

SHARED

FAVORITES

PRIVATE (8)

- List Public Tables
- BookReservations
- Loans
- Librarians
- Books
- Categories
- Authors
- Members

REFERENCE

- Templates
- Examples

```
1 CREATE TABLE Books (  
2   book_id INT PRIMARY KEY,  
3   title VARCHAR(100),  
4   author_id INT,  
5   category_id INT,  
6   publication_year INT,  
7   FOREIGN KEY (author_id) REFERENCES Authors(author_id),  
8   FOREIGN KEY (category_id) REFERENCES Categories(category_id)  
9 );
```

Results Explain Chart Export

Success. No rows returned

SQL Editor

Members Authors Categories Books Librarians Loans BookReservations +

Search queries... +

SHARED

FAVORITES

PRIVATE (7)

- BookReservations
- Loans
- Librarians
- Books
- Categories
- Authors
- Members

REFERENCE

- Templates
- Examples

```
1 CREATE TABLE Librarians (  
2   librarian_id INT PRIMARY KEY,  
3   librarian_name VARCHAR(100),  
4   email VARCHAR(100),  
5   shift VARCHAR(20)  
6 );
```

Results Explain Chart Export

Success. No rows returned

SQL Editor

Members Authors Categories Books Librarians Loans BookReservations List Public Tables +

Search queries... +

SHARED

FAVORITES

PRIVATE (8)

- List Public Tables
- BookReservations
- Loans
- Librarians
- Books
- Categories
- Authors
- Members

REFERENCE

- Templates
- Examples

```
1 CREATE TABLE Loans (  
2   loan_id INT PRIMARY KEY,  
3   member_id INT,  
4   book_id INT,  
5   librarian_id INT,  
6   loan_date DATE,  
7   return_date DATE,  
8   FOREIGN KEY (member_id) REFERENCES Members(member_id),  
9   FOREIGN KEY (book_id) REFERENCES Books(book_id),  
10  FOREIGN KEY (librarian_id) REFERENCES Librarians(librarian_id)  
11 );
```

Results Explain Chart Export

Success. No rows returned

Primary Database Role postgres Run

SQL Editor

Search queries...

+

SHARED

FAVORITES

PRIVATE (8)

List Public Tables

BookReservations

Loans

Librarians

Books

Categories

Authors

Members

REFERENCE

Templates

Examples

Members

Authors

Categories

Books

Librarians

Loans

BookReservations

List Public Tables

+

1

2

3

4

5

6

7

8

CREATE TABLE BookReservations (
 reservation_id INT PRIMARY KEY,
 member_id INT,
 book_id INT,
 reservation_date DATE,
 FOREIGN KEY (member_id) REFERENCES Members(member_id),
 FOREIGN KEY (book_id) REFERENCES Books(book_id)
);

Results

Explain

Chart

Export

Source

Primary Database

Success. No rows returned

SQL Editor

Search queries...

+

SHARED

FAVORITES

PRIVATE (8)

List Public Tables

BookReservations

Loans

Librarians

Books

Categories

Authors

Members

REFERENCE

Templates

Examples

Members

Authors

Categories

Books

Librarians

Loans

BookReservations

List Public Tables

+

1

2

3

SELECT table_name
FROM information_schema.tables
WHERE table_schema = 'public';

Results

Explain

Chart

Export

Source

Primary Database

table_name
members
librarians
categories
books
authors
loans
bookreservations

8

2.3 Table Design

Table	Primary Key
Members	member_id
Authors	author_id
Categories	category_id
Books	book_id
Librarians	librarian_id
Loans	loan_id
BookReservations	reservation_id

2.4 ERD



Relationship Explanation

A great many books can be penned by one author. Many books can be included under one category. A library member may check out numerous books. Many loans can be in the hands of one librarian. Reserving more than one book is allowed for one member.

2.5 Normalization (Third Normal Form)

Before Normalization At first, information about books may be registered as:

BookID	Title	AuthorName	CategoryName
--------	-------	------------	--------------

This structure introduces redundancy because author and category information are repeatedly stored.

After Normalization

The table was broken down into:

- Authors
- Categories
- Books

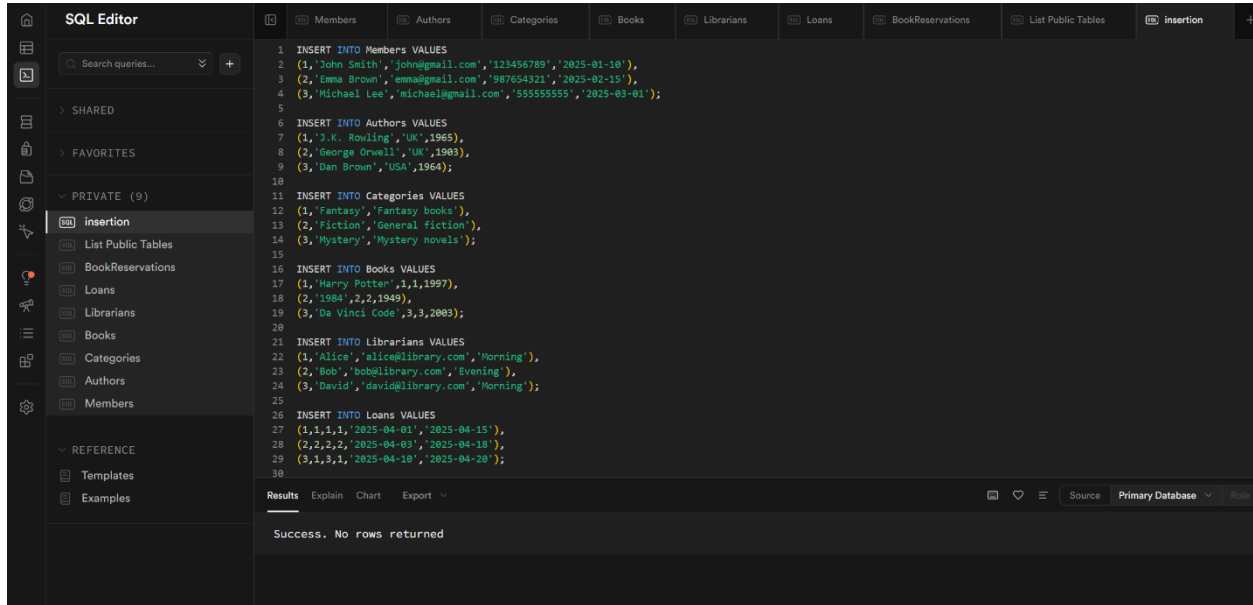
Benefits

- Eliminates duplicate data.
- Improves consistency.
- Reduces update anomalies.
- Simplifies maintenance.

3. Task 2 – Sample Data and Views

3.1 Sample Data

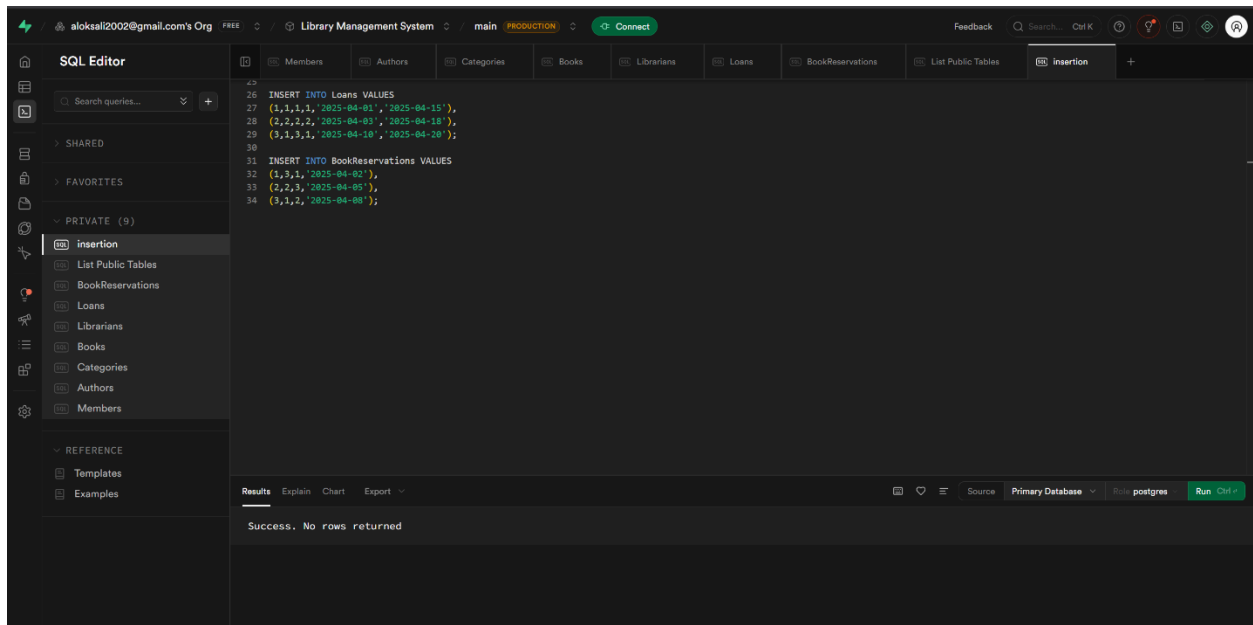
All the sample records were inserted in all the seven tables. All the 7 sample records were inserted into all seven tables. The data had a membership, books, authors, librarians, loans and reservations.



The screenshot shows a SQL Editor interface with a sidebar on the left containing a search bar and a list of tables: Members, Authors, Categories, Books, Librarians, Loans, BookReservations, List Public Tables, and Insertion. The main editor area displays a series of SQL INSERT statements for each table. The 'Insertion' table is currently selected in the sidebar. The statements are as follows:

```
1 INSERT INTO Members VALUES
2 (1, 'John Smith', 'john@gmail.com', '123456789', '2025-01-10'),
3 (2, 'Emma Brown', 'emma@gmail.com', '987654321', '2025-02-15'),
4 (3, 'Michael Lee', 'michael@gmail.com', '555555555', '2025-03-01');
5
6 INSERT INTO Authors VALUES
7 (1, 'J.K. Rowling', 'UK', 1965),
8 (2, 'George Orwell', 'UK', 1903),
9 (3, 'Dan Brown', 'USA', 1964);
10
11 INSERT INTO Categories VALUES
12 (1, 'Fantasy', 'Fantasy books'),
13 (2, 'Fiction', 'General fiction'),
14 (3, 'Mystery', 'Mystery novels');
15
16 INSERT INTO Books VALUES
17 (1, 'Harry Potter', 1, 1, 1997),
18 (2, '1984', 2, 2, 1949),
19 (3, 'Da Vinci Code', 3, 3, 2003);
20
21 INSERT INTO Librarians VALUES
22 (1, 'Alice', 'alice@library.com', 'Morning'),
23 (2, 'Bob', 'bob@library.com', 'Evening'),
24 (3, 'David', 'david@library.com', 'Morning');
25
26 INSERT INTO Loans VALUES
27 (1, 1, 1, '2025-04-01', '2025-04-15'),
28 (2, 2, 2, '2025-04-03', '2025-04-18'),
29 (3, 1, 3, '2025-04-10', '2025-04-20');
30
```

The bottom of the editor shows a 'Results' tab with the message: 'Success. No rows returned'.



This screenshot shows the same SQL Editor interface, but with the 'insertion' table selected in the sidebar. The main editor area displays the following SQL INSERT statements:

```
45 INSERT INTO Loans VALUES
46 (1, 1, 1, '2025-04-01', '2025-04-15'),
47 (2, 2, 2, '2025-04-03', '2025-04-18'),
48 (3, 1, 3, '2025-04-10', '2025-04-20');
49
50 INSERT INTO BookReservations VALUES
51 (1, 3, 1, '2025-04-02'),
52 (2, 2, 1, '2025-04-05'),
53 (3, 1, 2, '2025-04-08');
54
```

The 'Results' tab at the bottom again shows: 'Success. No rows returned'.

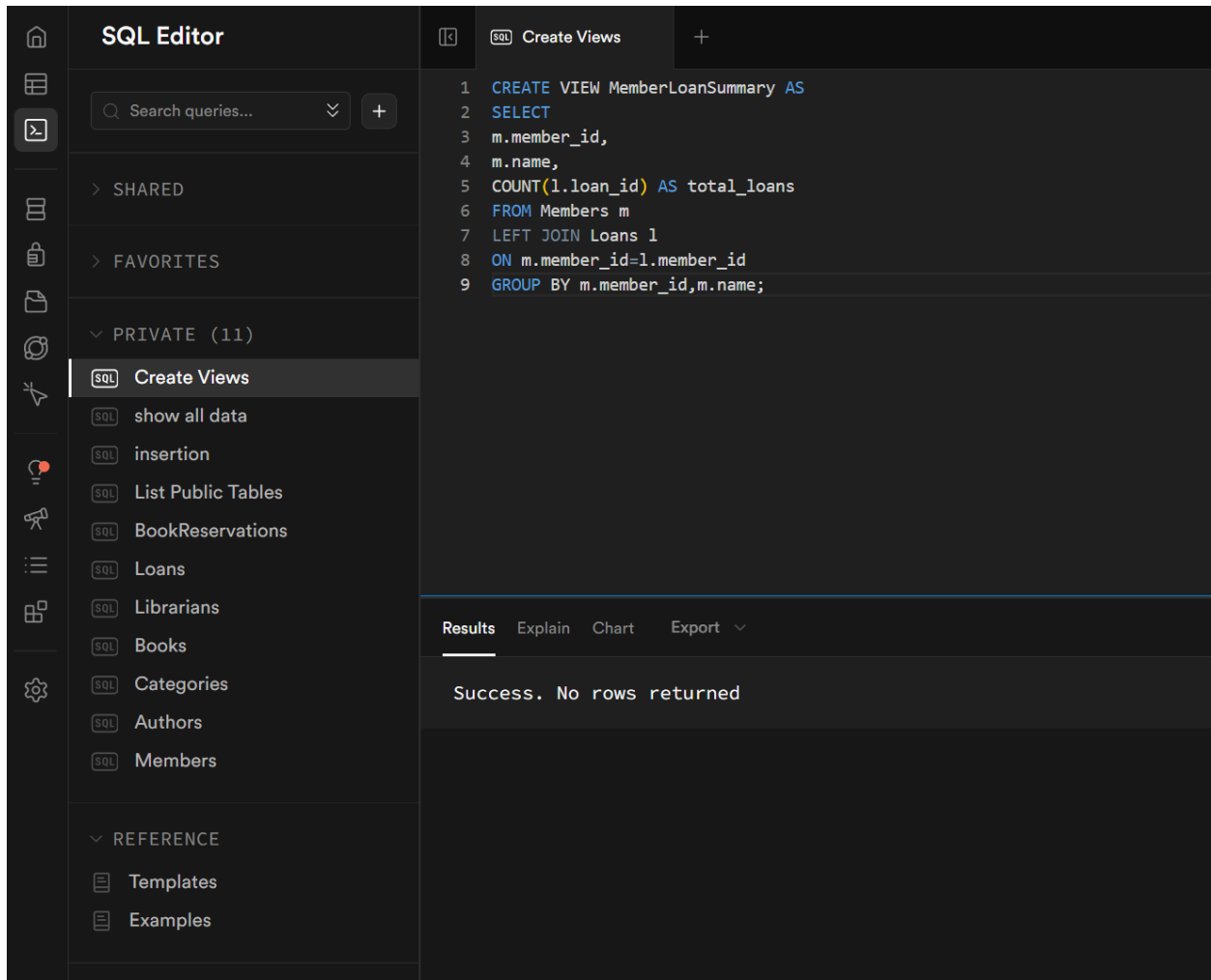
3.2 View 1 – Member Loan Summary

Purpose:

This perspective shows how many books each member borrowed.

Benefits

- Decides whether library users are active.
- Helps librarians keep track of loaner usage.



3.3 View 2 – Monthly Loan Trend

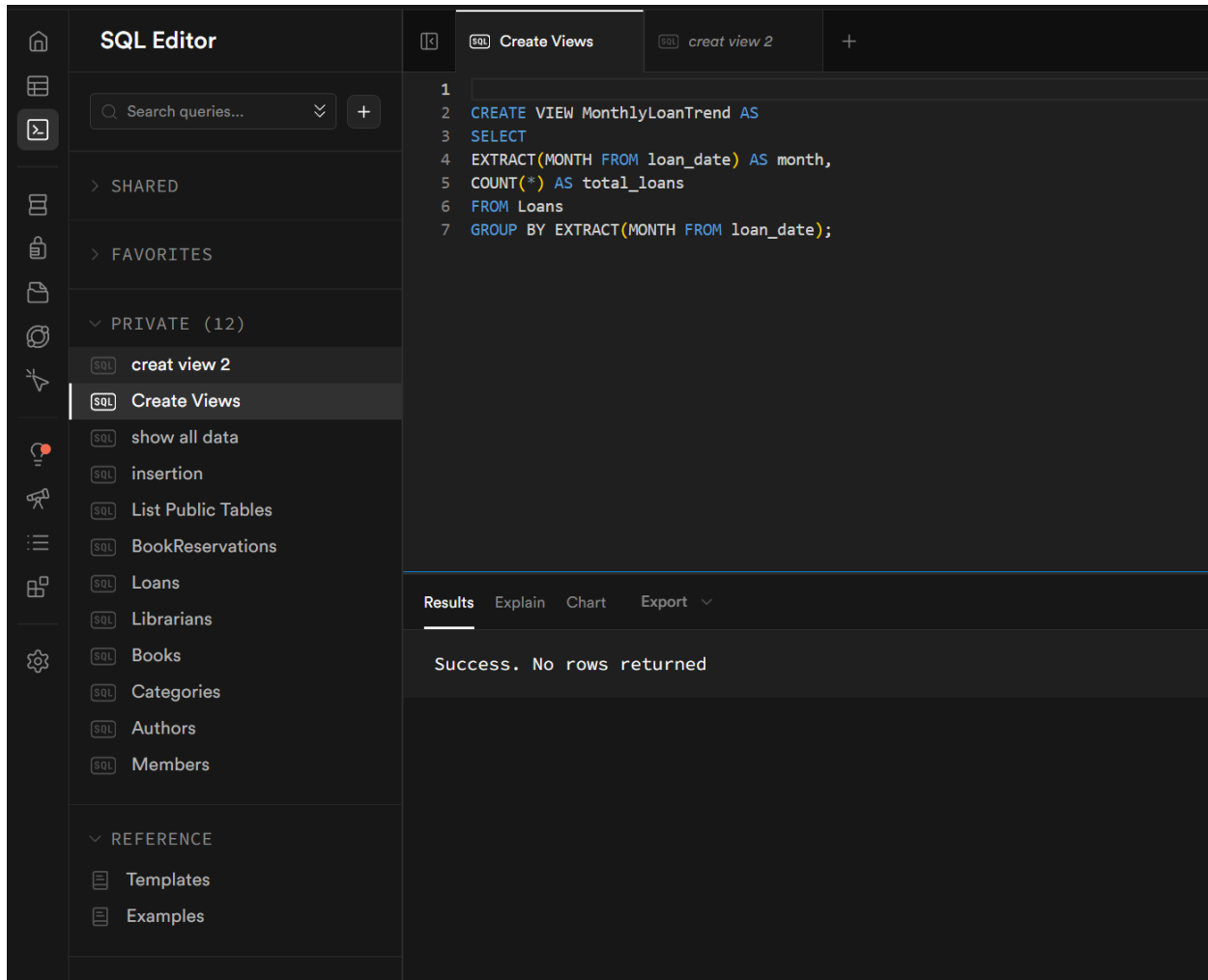
Purpose:

This will give you the total number of loans issued each month.

Benefits

Helps to determine borrowing patterns.

- Supports resource planning.



3.3 View 2 – Monthly Loan Trend

Purpose:

This view shows the number of loans issued during each month.

Benefits

- Helps identify borrowing trends.
- Supports resource planning.

The screenshot displays a SQL Editor interface with a dark theme. On the left, a sidebar contains a search bar and a list of queries categorized into 'SHARED', 'FAVORITES', and 'PRIVATE (12)'. The 'PRIVATE' section is expanded, showing a list of queries including 'creat view 2', 'Create Views', 'show all data', 'insertion', 'List Public Tables', 'BookReservations', 'Loans', 'Librarians', 'Books', 'Categories', 'Authors', and 'Members'. The 'Create Views' query is selected. The main editor area shows the SQL code for creating the 'MonthlyLoanTrend' view. The code is as follows:

```
1  
2 CREATE VIEW MonthlyLoanTrend AS  
3 SELECT  
4 EXTRACT(MONTH FROM loan_date) AS month,  
5 COUNT(*) AS total_loans  
6 FROM Loans  
7 GROUP BY EXTRACT(MONTH FROM loan_date);
```

Below the SQL editor, there are tabs for 'Results', 'Explain', 'Chart', and 'Export'. The 'Results' tab is active, displaying the message 'Success. No rows returned'.

4. Task 3 – SQL Querying and Analytics

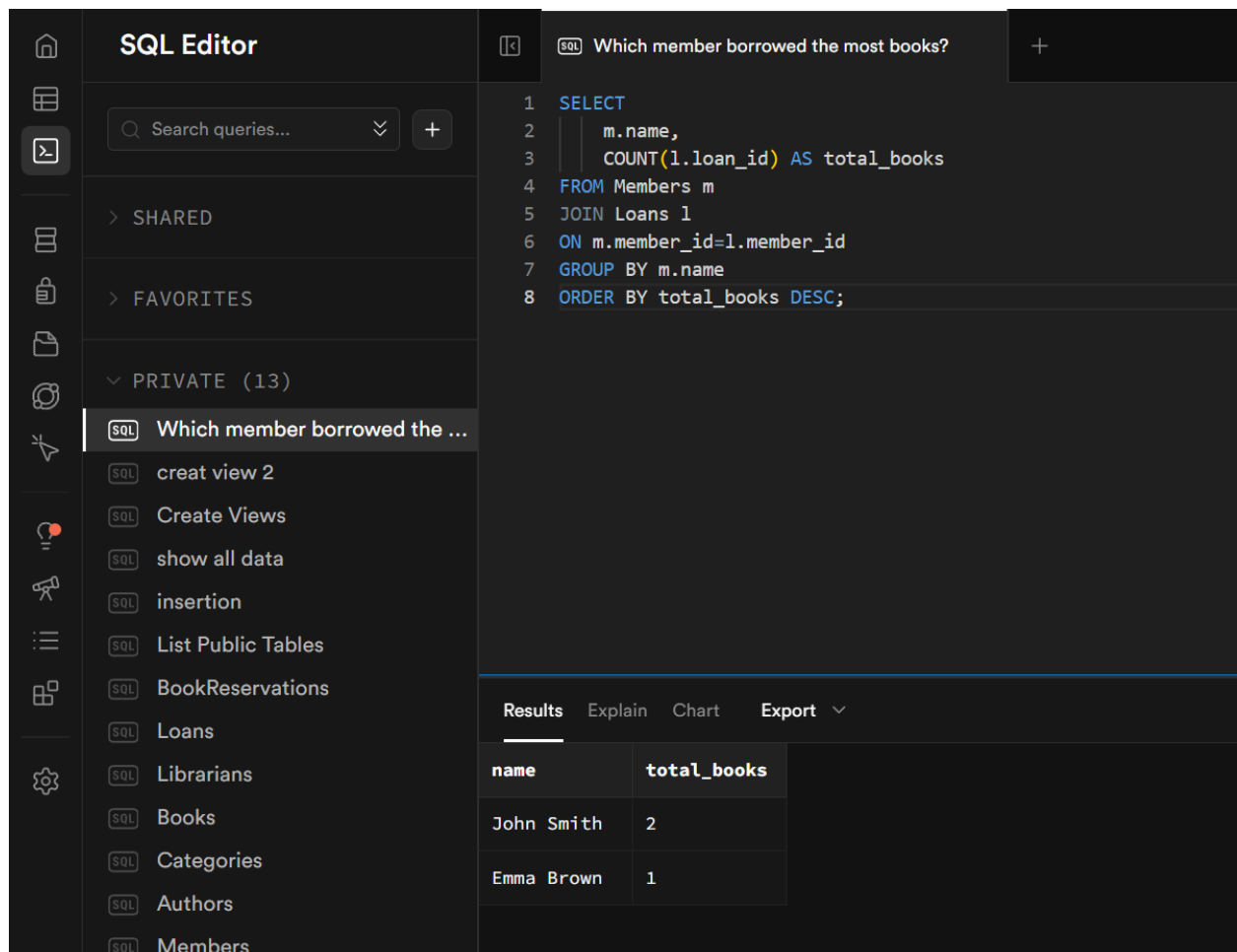
Query 1 – Member Borrowing Analysis

Explanation

The query joins Members and Loans tables and counts the number of books borrowed by each member.

Insight

The analysis reveals the most active library users.



The screenshot displays a SQL Editor interface with a dark theme. On the left is a sidebar with navigation icons and a list of queries. The main area shows an SQL query titled "Which member borrowed the most books?". Below the query editor, there are tabs for "Results", "Explain", "Chart", and "Export". The "Results" tab is active, showing a table with two columns: "name" and "total_books". The table contains two rows of data: "John Smith" with 2 books and "Emma Brown" with 1 book.

```
1 SELECT
2   m.name,
3   COUNT(l.loan_id) AS total_books
4 FROM Members m
5 JOIN Loans l
6 ON m.member_id=l.member_id
7 GROUP BY m.name
8 ORDER BY total_books DESC;
```

name	total_books
John Smith	2
Emma Brown	1

4. Task 3 – SQL Querying and Analytics

Query 1 – Member Borrowing Analysis

Explanation

A CASE statement classifies members into borrowing categories.

Insight

Members can be segmented based on activity levels.

The screenshot shows a SQL Editor interface with a sidebar on the left containing a search bar and a list of queries. The main editor displays an SQL query titled "Classify Members". The query uses a CASE statement to classify members into three categories: 'High User', 'Medium User', and 'Low User' based on their loan count. The query joins the 'Members' table with the 'Loans' table to count the number of loans per member. The results are displayed in a table with columns 'name' and 'member_type'.

```
1 SELECT
2   name,
3   CASE
4     WHEN loan_count >= 3 THEN 'High User'
5     WHEN loan_count >= 2 THEN 'Medium User'
6     ELSE 'Low User'
7   END AS member_type
8 FROM (
9   SELECT
10    m.name,
11    COUNT(l.loan_id) AS loan_count
12   FROM Members m
13   LEFT JOIN Loans l
14   ON m.member_id=l.member_id
15   GROUP BY m.name
16 ) x;
```

name	member_type
Emma Brown	Low User
John Smith	Medium User
Michael Lee	Low User

Query 3 – Book Ranking Analysis

Explanation

A window function ranks books according to the number of loans.

Insight

The ranking identifies the most popular books.

The screenshot displays a SQL Editor interface with a sidebar on the left containing a search bar and a list of queries and database objects. The main editor area shows a SQL query that ranks books by the number of loans. The results pane at the bottom shows a table with three columns: title, total_loans, and ranking. The results are ordered by total_loans in descending order, with 'Harry Potter' having 1 loan and a ranking of 1, 'Da Vinci Code' having 1 loan and a ranking of 1, and '1984' having 1 loan and a ranking of 1.

```
1
2 SELECT
3   b.title,
4   COUNT(l.loan_id) AS total_loans,
5   RANK() OVER(
6     ORDER BY COUNT(l.loan_id) DESC
7   ) AS ranking
8 FROM Books b
9 LEFT JOIN Loans l
10 ON b.book_id=l.book_id
11 GROUP BY b.title;
```

title	total_loans	ranking
Harry Potter	1	1
Da Vinci Code	1	1
1984	1	1

5. Task 4 – CAP Theorem

Consistency

Consistency means that everybody gets the same data, at the same time. Lending of books should, when loaned, clearly show that the book is on loan

Availability

Availability guarantees that the system will always be available to respond (in case of failures too).

Partition Tolerance

Partition tolerance is the ability to sustain service when an interruption happens between the servers.

The reason why All Three cannot be achieved simultaneously.

The CAP theorem states that distributed systems can only provide consistency, availability, and partition tolerance to one of these. When a network divides, the system has to satisfy two requirements—a consistency or an availability requirement.

To apply to the Library Management System.

The properties that are most important in a library management system are consistency and availability.

Consistency

Members should get right book availability.

Availability

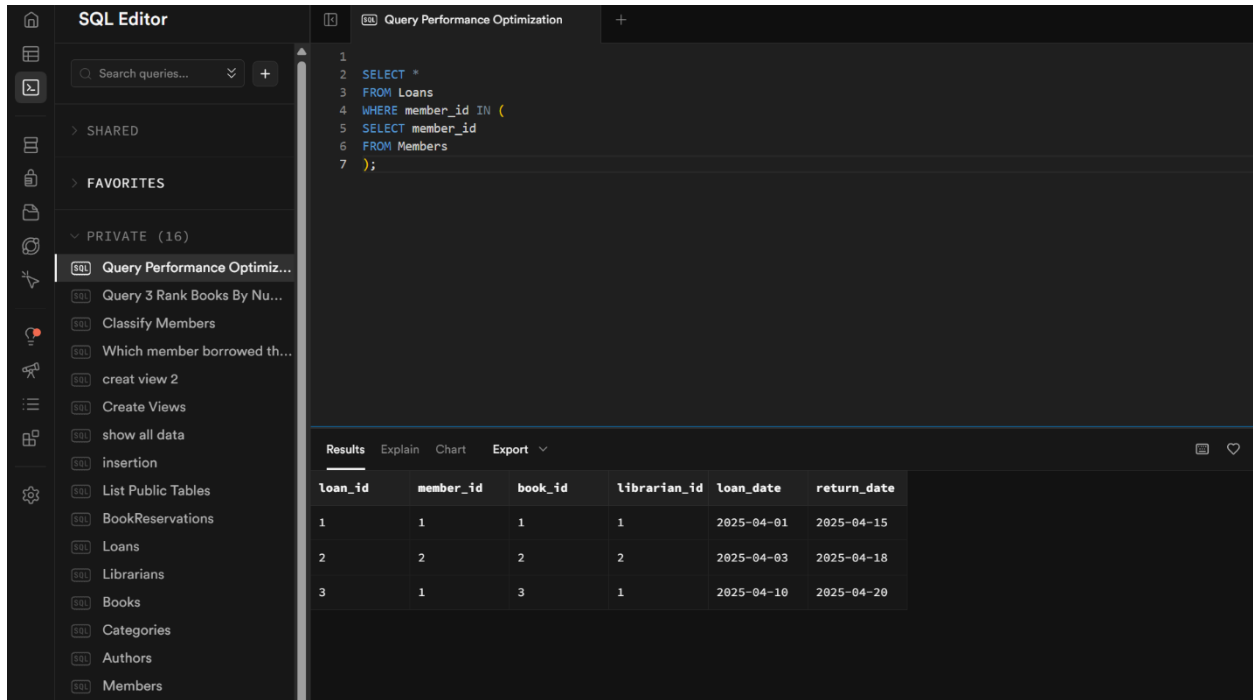
Users should be able to be able to search books and access services without interruptions.

Sacrificed Property

However, it is not as critical as partition tolerance because the system is typically operated within a centralized environment.

6. Task 5 – Query Performance Optimization

Original Query



The screenshot shows a SQL Editor interface with a sidebar on the left containing a search bar and a list of queries. The main editor displays a SQL query. Below the query, the 'Results' tab is active, showing a table with 6 columns: loan_id, member_id, book_id, librarian_id, loan_date, and return_date. The table contains 3 rows of data.

```
1 SELECT *
2 FROM Loans
3 WHERE member_id IN (
4   SELECT member_id
5   FROM Members
6 );
```

loan_id	member_id	book_id	librarian_id	loan_date	return_date
1	1	1	1	2025-04-01	2025-04-15
2	2	2	2	2025-04-03	2025-04-18
3	1	3	1	2025-04-10	2025-04-20

Issues

- Uses unnecessary subqueries.
- Becomes inefficient for large datasets.

Optimized Query

The screenshot displays a SQL Editor window with a sidebar on the left containing a search bar and a list of queries. The main editor area shows the following SQL code:

```
1 CREATE INDEX idx_member
2 ON Loans(member_id);
3
4 SELECT *
5 FROM Loans
6 JOIN Members
7 ON Loans.member_id=Members.member_id;
```

Below the code, the 'Results' tab is active, showing a table with 10 columns: loan_id, member_id, book_id, librarian_id, loan_date, return_date, name, email, phone, and join_date. The table contains three rows of data.

loan_id	member_id	book_id	librarian_id	loan_date	return_date	name	email	phone	join_date
1	1	1	1	2025-04-01	2025-04-15	John Smith	john@gmail.com	123456789	2025-01-10
2	2	2	2	2025-04-03	2025-04-18	Emma Brown	emma@gmail.com	987654321	2025-02-15
3	1	3	1	2025-04-10	2025-04-20	John Smith	john@gmail.com	123456789	2025-01-10

Improvement

The optimized version relies on Index, and a join operation. This will help the execution time and scalability issues to increase if the database grows.

7. Conclusion

The project has been designed and implemented successfully the relational database for Library Management System project. A set of seven normalized tables was created and they were then joined together with a primary and foreign key. Views and SQL queries are employed as a means to sample and analyze. The analytical queries showed how useful the database system is in providing value to the business. The concepts of CAP-theorem were tested in the context of the library environment and methods of query optimization were employed to improve the performance. In general, the assignment demonstrates the significance of database design, database normalization, SQL analytics, database performance tuning, and using databases in modern data management systems.

8. References

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7. Appendix

<https://github.com/jimsonjames007-debug/Library-Management-System.git>